MINOR PROJECT – 2

PROJECT INTRODUCTION

Project Title
Diabetes Prediction Using Logistic Regression

Name
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Dataset Name
Pima Indians Diabetes Dataset

Dataset Link
https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

Short Description of the Dataset
The dataset contains medical records used to predict whether a person is diabetic or non-diabetic. It includes 8 key health attributes describing each individual: number of pregnancies, plasma glucose concentration, diastolic blood pressure, triceps skin-fold thickness, serum insulin level, Body Mass Index (BMI), diabetes pedigree function, and age. The target variable indicates diabetes status (0 for non-diabetic, 1 for diabetic).

Objective of the Analysis
The objective of this project is to develop a Logistic Regression model to predict diabetes based on patient medical data. The work includes checking and handling missing values, detecting and treating outliers using the IQR method, standardizing features, splitting the dataset into training and testing sets, training the classification model, and evaluating performance using accuracy, precision, recall, F1-score, ROC-AUC score, confusion matrix, and ROC curves.